Quantum Reference Algebra for Information Flow

World structure and physical control

Complete Collected Volume

Sidong Liu, PhD

iBioStratix Ltd

sidongliu@hotmail.com

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Contents

The numbers below refer to pages in this PDF. Each paper retains its own printed pagination, main argument and references. The technical appendices of Q-RAIF C are distributed as separate linked supplements; the other papers retain their appendices.

Introduction and reading map 3

QR-A Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass 4

QR-B Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost 28

QR-C Representing Control in the World: Retained History, Physical Sources and Renewal in Q-RAIF 50

Introduction

Why should the algebra through which a system acts be related to the structure through which its world propagates? Q-RAIF studies this interface directly. A positive control cost, an information metric and a Lorentzian form are distinct objects; a bridge has to follow the actual generators and their representation.

A derives the common quadratic shell and Clifford relations under its local propagation assumptions. B shows when a centred field-pulse cost is independent of the unknown carrier state, and when generator alignment relates that cost to the propagation form. C contrasts control discarded into a continuous-spectrum port with control retained in the closed propagation. The former restricts the exact multiplicative domain; the latter permits an exact represented algebra and a conditional Lorentzian extension.

Finite field actuators, material sources, clocks and receiver renewal provide physical witnesses and resource tests for these interfaces. They do not prove arbitrary world–control alignment or indefinitely cyclic operation. Mass–Energy, Null Cluster and No View remain separate companion papers: positive spectral readout, a matched null-access task and conservation-constrained source selection complete the six-paper responsibility without changing this volume's three-paper membership.

Reading map

QR-A Local propagation and the single quadratic shell constrain Clifford and Lorentzian readout. A common cone still needs the stated speed-alignment dynamics.

QR-B A centred physical coupling gives a state-independent cost criterion and a conditional bridge to the propagation form. Alignment is an additional physical question.

QR-C Discarded control ports and closed propagation give different exact representation results. Finite source, clock and renewal witnesses test their physical implementation.